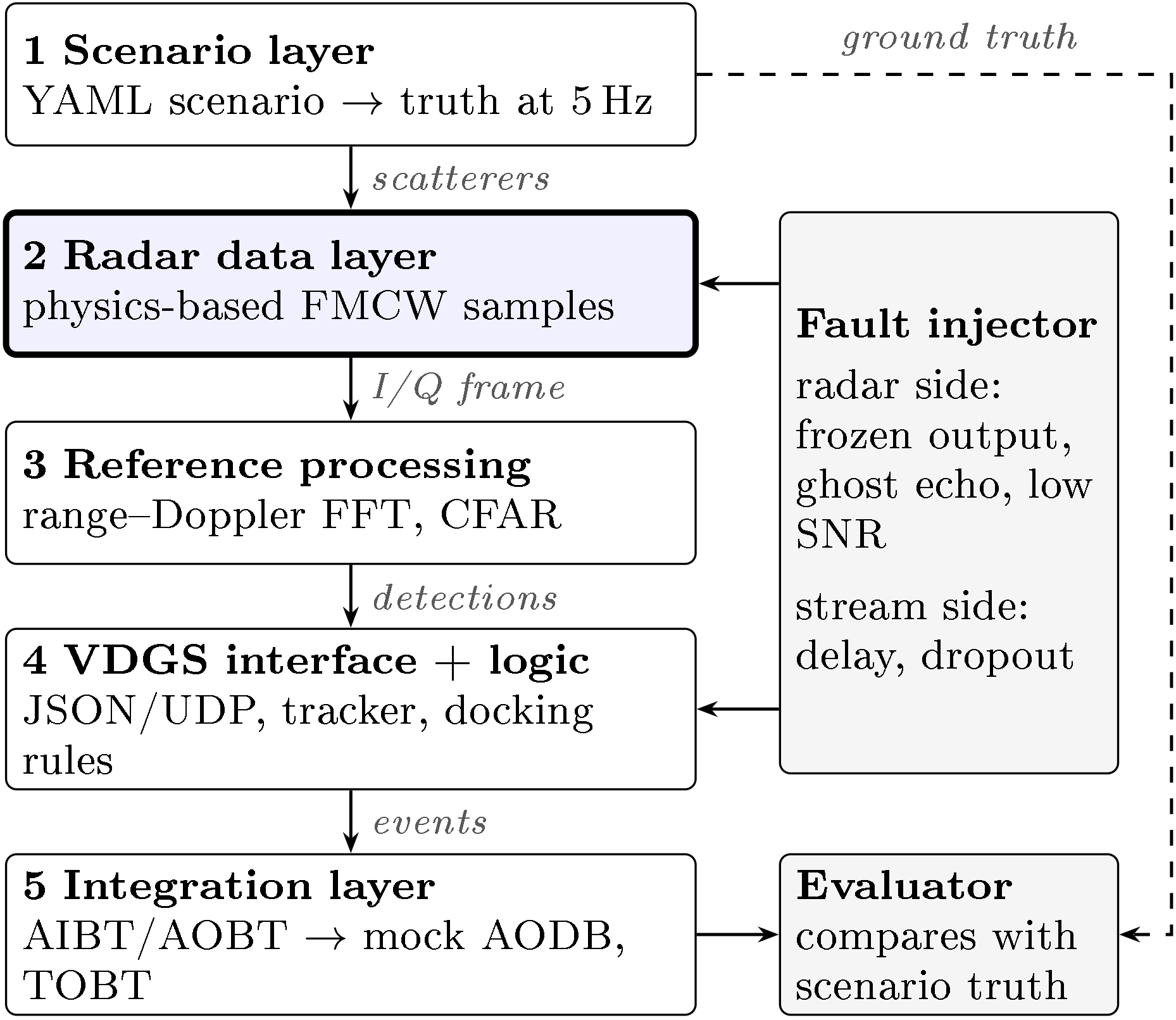
1 Scenario layer
YAML scenario → truth at 5 Hz
ground truth
scatterers
2 Radar data layer
physics-based FMCW samples
I/Q frame
3 Reference processing
range–Doppler FFT, CFAR
detections
4 VDGS interface + logic
JSON/UDP, tracker, docking rules
events
5 Integration layer
AIBT/AOBT → mock AODB, TOBT
Fault injector
radar side: frozen output, ghost echo, low SNR
stream side: delay, dropout
Evaluator
compares with scenario truth